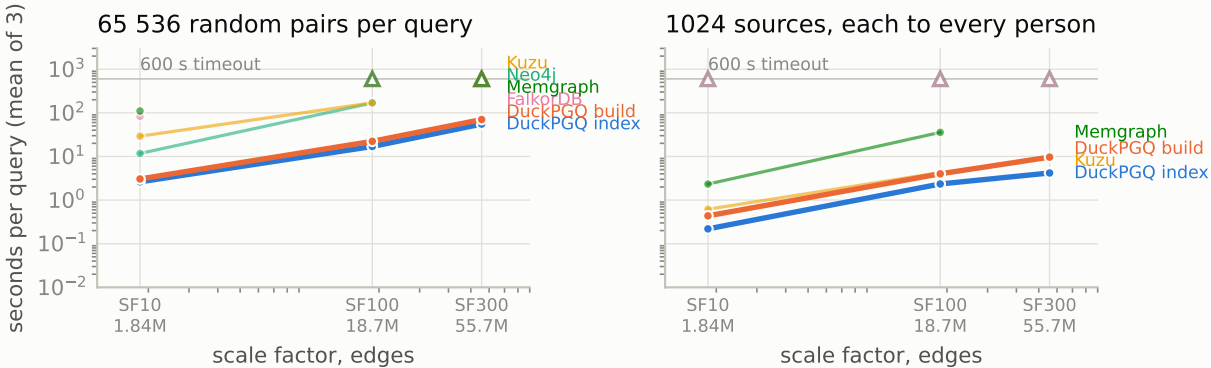Scaling with the graph: the largest point of each shape from SF10 to SF300